D119-D120 | Counterfactual before contribution

PRIVATE FICTIONAL teaching only. No actual loans, users, sellers, subsidy, experiment, dataset, rollout, payment or real decision authority.

Attempt without notes	D119 is a fresh mixed retrieval case, not an established weakest-skill diagnosis. Write actual response and confidence only if performed. Review methods after attempting; delayed changed-context retest is not recorded as done.
D119 prediction	800 locked scores .05 and 200 at .30 predict 100 defaults. New labels 80/80 give empirical 10%/40%, 160 defaults and O/E 1.6. Do not fit to these labels and call them an untouched final test.
D119 assignment	A 150/500 and B165/500 give ITT +3 pp; exposed A150/300 and B165/450 give -13.333333 pp. Exposure is postassignment; the sign reversal does not identify a causal treatment-on-treated effect.
D119 seller economics	Treated 120->170, comparison 100->130, 50 sellers: arithmetic DiD 20 per seller, conditional incremental GMV 1000, net take 8% minus subsidy 100 yields -20. GMV is not profit and four means cannot supply uncertainty.
D120 counterfactual and economics	Treated 1000->1500/control 800->1040: counterfactual 1240, DiD 260 per seller, total 26000. Net 6% yields 1560 before subsidy 2000/setup 300, contribution -740. No automatic causal or rollout conclusion.
Sensitivity and contradiction	gamma is treated-minus-control untreated post change: candidate effect 260-gamma. Assumed[-100,100] gives effects[160,360] and contribution[-1340,-140]. Credibly control-only promotion 80 gives adjusted effect 340 and contribution -260. Neither is a statistical interval or permission to discard inconvenient data.
Limits	All prose is Case-v1, revise after edits. Native arithmetic is not empirical identification, statistical uncertainty, new source/full AV/cross-viewer/independent learner/account/legal clearance or overall10. Actual learner/reviewer/delayed fields remain blank.

D119 inputs | Three independent fresh cases

Counts are integers, complete binary outcomes. Prices/GMV are rupees per fixed equal-month seller, net take is after variable cost. No actual data or implied authority.

Low-bin loans	800.00
Low-bin probability	0.05
Low-bin actual defaults	80.00
High-bin loans	200.00
High-bin probability	0.30
High-bin actual defaults	80.00
Arm A assigned	500.00
Arm A exposed	300.00
Arm A activated	150.00
Arm B assigned	500.00
Arm B exposed	450.00
Arm B activated	165.00
Treated pre GMV per seller	120.00
Treated post GMV per seller	170.00
Comparison pre GMV per seller	100.00
Comparison post GMV per seller	130.00
Treated fixed seller count	50.00
Net platform take fraction	0.08
Total cohort subsidy	100.00

Structural inputs valid? (1=yes)	1.000000

Yellow cells are fictional assumptions, not observed data or permission. A structural pass does not establish current source definitions, demand, contractual terms, economic feasibility, liquidity or real authority.

D119 answers | Denominators can reverse a contrast

All three cases remain separate. ITT uses assigned counts, exposure rates are descriptive conditional contrasts. Conditional seller effect assumes untreated parallel changes and no differential shock/interference.

Measure	Value
Expected defaults	100.000000
Actual defaults	160.000000
Low-bin empirical default fraction	10.0000%
High-bin empirical default fraction	40.0000%
Observed / expected defaults	1.600000
Arm A assigned-user activation	30.0000%
Arm B assigned-user activation	33.0000%
ITT difference B minus A	3.0000%
Arm A exposed-user activation	50.0000%
Arm B exposed-user activation	36.6667%
Exposure-conditioned difference	-13.3333%
Comparison GMV change per seller	30.000000
Conditional treated counterfactual	150.000000
Arithmetic DiD per seller	20.000000
Conditional incremental cohort GMV	1000.000000
Conditional incremental contribution	-20.000000

Correct only B activated165->145: ITT becomes -1 percentage point, not a changed assignment count. No CI or causal identification inferred from these aggregates. Actual error selection and delayed retest are blank.

D120 inputs | Levels, differences and costs

Fixed treated membership over matched one-month periods; GMV is seller sales, not platform revenue. Commission and variable cost are fractions of GMV; subsidy/setup are incremental cohort totals.

Treated pre GMV per seller	1000.00
Treated post GMV per seller	1500.00
Comparison pre GMV per seller	800.00
Comparison post GMV per seller	1040.00
Number of fixed treated sellers	100.00
Platform commission fraction	0.10
Variable service cost fraction	0.04
Total fixed cohort subsidy	2000.00
Incremental cohort setup cost	300.00
Older treated pre GMV per seller	900.00
Older comparison pre GMV per seller	700.00
Assumed gamma lower INR per seller	-100.00
Assumed gamma upper INR per seller	100.00
Credibly separate control-only promotion	80.00

Structural inputs valid? (1=yes)	1.000000

Yellow cells are fictional assumptions, not observed data or permission. A structural pass does not establish current source definitions, demand, contractual terms, economic feasibility, liquidity or real authority.

D120 counterfactual | Arithmetic needs assumptions

Conditional untreated parallel changes, no anticipation/spillover, fixed composition and no concurrent differential shock. These assumptions are not established by the supplied means.

Measure	Value
Raw post treated-minus-comparison gap	460.000000
Treated observed before-after change	500.000000
Comparison observed before-after change	240.000000
Conditional treated post counterfactual	1240.000000
Arithmetic DiD per treated seller	260.000000
Conditional incremental treated cohort GMV	26000.000000

Earlier treated pre change	100.000000
Earlier comparison pre change	100.000000
Earlier treated-minus-comparison change	0.000000

Equal earlier +100/+100 changes are compatible with the assumption, not proof of the unobserved post path. Nonsignificant pretrend tests may lack power; four means and seller counts cannot yield a defensible statistical confidence interval.

D120 economics | GMV is not platform contribution

Rupees over the one-month cohort. Net margin is commission minus variable expense. Fixed costs are incremental once only. Negative subsidy bound is not a permitted negative offer.

Measure	Value
Net incremental platform margin fraction	6.0000%
Conditional incremental cohort GMV	26000.000000
Incremental commission revenue	2600.000000
Incremental variable service expense	1040.000000
Incremental margin before fixed costs	1560.000000
Total incremental fixed costs	2300.000000
Conditional incremental contribution	-740.000000
Required expected GMV at positive net margin	38333.333333
Minimum whole-rupee required GMV	38334.000000
Max arithmetic subsidy, may be negative	1260.000000
Arithmetic subsidy bound per seller	12.600000
Actual total treated post GMV LEVEL	150000.000000
Total treated commission LEVEL	15000.000000
Total treated variable expense LEVEL	6000.000000
Wrong raw-change margin, shown for comparison	3000.000000

Baseline contribution -740. Required whole GMV 38334 is an arithmetic threshold, not available demand. Max subsidy 1260 or 12.60 per seller conditional on unchanged behavior. Changing subsidy can change behavior; neither threshold is a robust recommendation or authorization.

D120 sensitivity | Assumptions, not a confidence set

gamma = treated untreated post change MINUS comparison untreated post change. Candidate effect = observed DiD - gamma. Bounds are explicitly chosen assumptions, not estimated uncertainty.

Measure	Value
Assumed gamma lower INR per seller	-100.000000
Assumed gamma upper INR per seller	100.000000
Effect lower: DiD minus gamma upper	160.000000
Effect upper: DiD minus gamma lower	360.000000
Conditional lower incremental GMV	16000.000000
Conditional upper incremental GMV	36000.000000
Contribution at lower effect endpoint	-1340.000000
Contribution at upper effect endpoint	-140.000000
Contribution minimum, handles negative net margin	-1340.000000
Contribution maximum, handles negative net margin	-140.000000

Baseline gamma [-100,100] implies effects[160,360] and contribution[-1340,-140]. Sort economic endpoints if net margin changes sign. A different range may reverse the sign; no universal causal effect, statistical interval, HonestDiD implementation or real action follows.

D120 objection | A correction needs separate evidence

Control promotion 80 is stipulated independently attributable only to comparison sellers, without interaction, treated spillover or preperiod effect. This is an illustrative conditional correction.

Measure	Value
Observed comparison post GMV	1040.000000
Separately attributed control-only promotion	80.000000
Adjusted comparison post without promotion	960.000000
Adjusted comparison change	160.000000
Conditional treated counterfactual	1160.000000
Conditional adjusted DiD per seller	340.000000
Conditional adjusted incremental cohort GMV	34000.000000
Conditional adjusted contribution	-260.000000

Baseline adjusted effect 340/cohort34000/contribution -260. It is not acceptable to remove inconvenient controls without evidence. If attribution, interaction, spillover or timing differs, this simple correction fails. No actual promotion/source evidence is supplied.

Audit | Arithmetic is not causal evidence

Independent retrieval and seller cases. Invalid assumptions and broken result references hold affected summaries; no real acceptance or approval.

Retrieval identities valid? (1=yes)	1.000000
Guarded fresh retrieval contribution	-20.000000
Seller identities valid? (1=yes)	1.000000
Guarded baseline seller contribution	-740.000000

A passing flag checks specified identities only. Four aggregate means cannot establish assignment, parallel counterfactual, seller-level uncertainty, current terms, feasibility or authority. Real action remains stopped.

Evidence | Counterfactual and arithmetic differ

Original fictional D119/D120 exercises and existing selected readings retained at https://top1os.dev/. No new empirical reading or actual seller/learner observation is claimed.

Lineage	Retrieval inputs -> Retrieval -> Audit. Seller inputs -> Counterfactual -> Economics/Sensitivity/Promotion objection -> Audit. Each rupee figure is matched to cohort and one-month window.
Strongest identification objection	Comparison membership, shocks, spillovers, anticipation, selection and time-varying composition need evidence. Equal prechanges or a nonrejected test cannot prove the missing untreated post outcome.
Strongest economic objection	Commission revenue and GMV levels are not incremental contribution. Net take must exclude the stated variable expense exactly once; subsidy/setup are incremental, not silently duplicated. Changing the offer can change effect.
Inference not fabricated	Seller counts and means do not reveal seller variation, repeated-unit dependence, assignment or clusters. No p-value, standard error, confidence set or statistical significance is created.
Fictional day0/day30 hold	Data/causal/finance roles would need original evidence, definitions and allowed audience before any real decision. No real person accepted a role. No seller contact, experiment, payment, data export, disclosure or spend.
Delayed fresh transfer	After an actual attempt choose provisional remediation from new errors. On a delayed day change rates, denominators, timing, gamma and costs; record actual response/objection/correction only if performed. No measured weakest skills or mastery inferred.

Actual work | Not a simulated learner record

Actual attempts, confidence, observed errors, review objections, correction effects and delayed transfer remain blank until performed with authority.

Actual date / case	Attempt / confidence	Input / evidence	Observed error / reason	Correction / effect	Objection / reply	Delayed context / result

Actual work remains blank. Record only what occurred: dates, reasoning, error/source/correction, strongest objection and a new delayed transfer. No invented learner results, sponsor acceptance, actual budget release or signed rights.

Failure log | Authored execution only

Observed native tests are added only after execution. Not actual learner mistakes, independent human audit, Excel/Sheets parity, real authority or overall 10/10.

Controlled test	Observed authored outcome
Baseline independent retrieval and seller formulas	PASS: every baseline formula independently recomputed.
Corrected activation / zero exposures / scores	PASS: changed activation, labels, zero scores/exposure.
Seller count / GMV / cost changes	PASS: seller counts, zero/negative effect and commission/costs.
Positive / zero / negative net margin	PASS: positive, zero, negative net margins; thresholds hold.
Asymmetric gamma / reversed economics endpoints	PASS: asymmetric gamma and economic endpoint sorting.
Promotion zero / changed / boundary	PASS: control-only promotion zero/changed/boundary.
Invalid inputs / independent-case isolation	PASS:16 invalid cases block affected output, independent case preserved.
Broken refs / affected Audit holds	PASS:5 broken refs hold affected Audit, other case preserved.
Restored baseline / actual work blank	PASS:baseline restored; actual work blank, no learner claims.

Only the original synthetic course exercise at https://top1os.dev/ supplies the case figures. No new empirical reading, real participants, company records, Indian legal clearance, actual commitments or source completeness claimed.